

When Sample Selection Bias Precipitates Model Collapse

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Introduction

Recursive synthetic-data training can narrow distributional support when later generations are selected by biased local references. The question is when accumulation preserves diversity.

Theory

Replace paradigm [R1]

Latest-generation replacement drives covariance loss and mode narrowing.

Accumulate paradigm [R2]

Pooling generations can stabilize unbiased recursion, but biased selection changes the limit.

Theorem 1

Top-alpha selection moves the mean toward the local reference while covariance contracts.

Theorem 2

Covariance trace follows a power-law decay in the local basin.

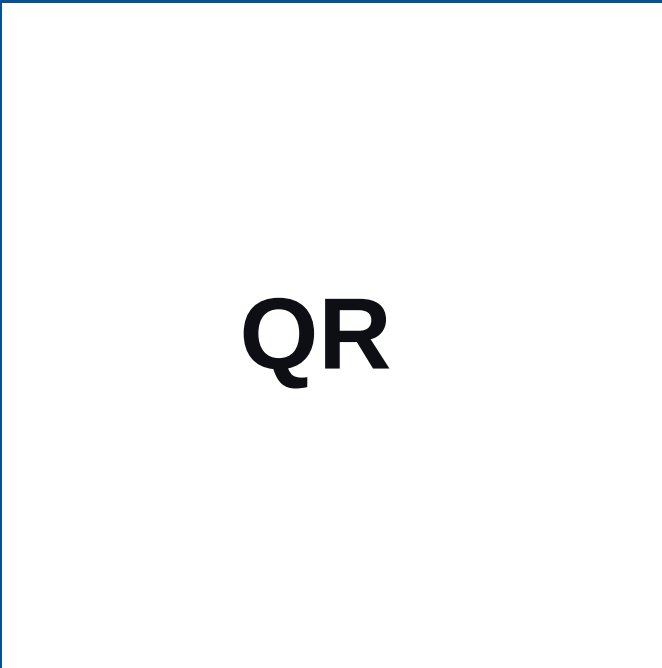
Theorem 3

After fitting error is small, target performance is governed by distribution mismatch.

References

[R1] Replace-paradigm collapse analysis. [R2] Accumulate-paradigm recursive training analysis.

For more methodology and theoretical details, please refer to the full paper.



Scan

OpenReview paper and supplement.

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Selection bias
accelerates collapse.
Accumulation slows
but does not prevent it.
Scaling sample count
does not restore diversity.

Multivariate Gaussian Modeling

Gaussian

Selection contracts covariance under replacement and accumulation.

Anisotropic Gaussian

Directional variance changes the path but not dispersion loss.

Gaussian mixture

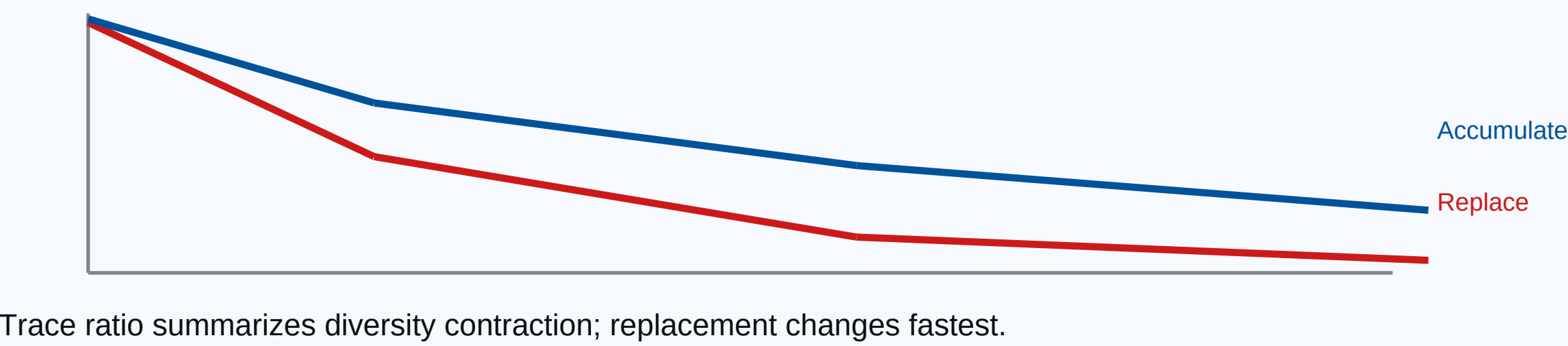
Biased filtering reduces mode and tail coverage.

Laplace

Heavy-tailed initialization still loses dispersion.

Collapse Signal

Trace-ratio diagnostic



Empirical Summary

Across Gaussian and non-Gaussian families, biased selection is the dominant driver of diversity loss and reduced coverage.

For additional experimental results, please refer to the full paper.

